
Refinement Steps Buy Intelligibility, Not Identity: Metric-Selective Test-Time Scaling in Masked-Diffusion TTS

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Abstract

1 A diffusion language model buys sequential computation twice: depth, paid once
2 in parameters, and denoising steps, rented per utterance at serve time. We ask
3 whether the rented kind helps every capability equally. We train an iso-parameter
4 width–depth grid of 15 masked-diffusion codec TTS models (3 seeds, 20–125 M
5 non-embedding parameters) and sweep the refinement budget T at inference only,
6 giving a 75-point (w, d, T) surface per metric from 45 trained models. Fitting $\text{err} =$
7 $E + Aw^{-\alpha} + Bd^{-\beta} + CT^{-\tau}$ pre-registered, refinement steps attack intelligibility
8 harder than speaker identity: $\Delta\tau = \tau_{\text{WER}} - \tau_{\text{SIM}} = 0.110$, 95% run-level bootstrap
9 CI [0.092, 0.131]. Depth and steps are *not* interchangeable: the substitution form
10 $B(dT^\kappa)^{-\beta}$ is worse than the separable form by $\Delta\text{AICc} = +69.3$. Steps are an
11 intelligibility knob, not a general one.

1 Introduction

13 A diffusion language model can buy sequential computation twice. It *owns* depth: d transformer
14 blocks, paid once in parameters and then on every forward pass forever. It *rents* refinement steps:
15 T denoising passes over the same sequence, paid per utterance and adjustable at serve time. An
16 autoregressive model has only the first lever; its serial cost per token is fixed by the architecture the
17 moment training ends. If rented serial compute substitutes for owned serial compute, a deployment
18 can trade parameters for steps after the fact — but only for the capabilities that steps actually improve.

19 Zero-shot text-to-speech is an unusually clean probe of that question, because it separates two
20 capabilities that are measured by standard, independent metrics. Intelligibility is compositional: the
21 model must place the right phonetic content in the right order, and word error rate (WER) reads it
22 out. Speaker identity is representational: the model must carry timbre from a short prompt into new
23 content, and speaker similarity (SIM-o) reads that out. The two need not respond to the same knob.

24 We train one iso-parameter width–depth grid of masked-diffusion codec language models and then
25 sweep T at inference only, which yields a full (w, d, T) response surface per metric at zero additional
26 training cost. Two claims were pre-registered before any run. **H-D2**: refinement steps improve
27 intelligibility more than identity, i.e. test-time scaling is metric-selective. **H-D3**: for intelligibility,
28 steps and depth exchange at a measurable rate κ — how many step-doublings buy one depth-doubling.

$$\text{err}_m(w, d, T) = E_m + A_m w^{-\alpha_m} + B_m d^{-\beta_m} + C_m T^{-\tau_m} \quad \text{err}_m(w, d, T) = E_m + A_m w^{-\alpha_m} + B_m (dT^{\kappa_m})^{-\beta_m} \quad (1)$$

29 Both claims were registered before the first run, together with the thresholds, the frozen sampler
30 and the analysis plan. The answer to the first is yes and quantitative: $\Delta\tau = 0.110$ with a run-level
31 bootstrap CI of [0.092, 0.131], so the step exponent is reliably larger for intelligibility than for identity.

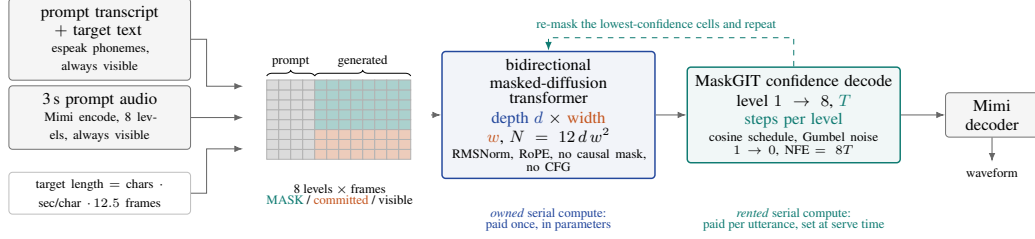


Figure 1: *The two ways a masked-diffusion TTS model buys sequential computation.* Phonemes and a 3-second prompt are always visible; the generated region of the $8 \times \text{frames}$ token grid starts fully masked. The backbone spends *depth* — *owned* serial compute, fixed at training time — and the sampler spends *T refinement steps per codebook level* — *rented* serial compute, chosen at serve time. One training grid over (w, d) therefore yields a whole (w, d, T) surface at inference. This paper asks whether the rented kind helps every capability equally.

The answer to the second is a clean no: the substitution form is decisively *worse* than the separable one ($\Delta\text{AICc} = +69.3$ for WER, $+71.3$ for SIM-o) and κ runs to its bound, so no finite depth-step exchange rate describes this regime. The two results together say something sharper than either alone: extra inference compute is a targeted instrument, and it cannot be treated as a substitute for architecture.

- ❶ The first per-metric (w, d, T) scaling surface for speech generation: 75 points per metric from one training grid of 45 models, released with the harness.
- ❷ $\Delta\tau = 0.110$, CI $[0.092, 0.131]$ — steps are metric-selective — and the *negative* result on depth-step substitution stated as an AICc comparison ($+69.3$) rather than a narrative, with κ reported as unidentified.
- ❸ A serving rule measured rather than assumed: c_{layer} is flat in *width* at batch 1 on the evaluation GPU, so serial latency is $8Td_{\text{layer}}$ and the *step budget* is the only latency knob that trades against intelligibility.

2 Related work

Masked diffusion language models are now trained at scale [Nie et al., 2025, Sahoo et al., 2024, Austin et al., 2021], but their scaling is reported against a single likelihood-shaped objective, so a per-capability question cannot arise. In speech, the masked-generative recipes that made codec-token TTS practical — confidence-decoded parallel generation over residual codebook levels [Borsos et al., 2023, Chang et al., 2022] and its multi-stage successors [Wang et al., 2024] — fixed a step budget by engineering judgement and reported a single operating point; none maps error against steps as a law. Autoregressive codec TTS [Wang et al., 2023] has begun to report scaling behaviour, but in total parameters N only [Ye et al., 2025], which cannot separate width from depth and has no step axis at all; classical scaling work is likewise N -indexed [Kaplan et al., 2020, Hoffmann et al., 2022]. Test-time scaling is well studied in reasoning [Snell et al., 2024], where the payoff is measured on one correctness metric; whether extra inference compute helps different capabilities *differently* is, to our knowledge, untested in generative speech. That is the gap this abstract fills: per-metric laws in (w, d, T) for a masked-diffusion TTS model.

3 Method

Backbone and objective. A bidirectional (non-causal) pre-LN transformer over Mimi [Défossez et al., 2024] codec tokens at 12.5 Hz with the first 8 residual codebooks (2048 entries each, plus MASK and PAD). *Width* w , *depth* d , head dimension 64, GELU MLP ratio 4, RMSNorm, RoPE, no weight tying, so the non-embedding parameter count is $N = 12dw^2$. One embedding table per codebook is summed at the input and one linear head per codebook reads out. Training is SoundStorm-style [Borsos et al., 2023] coarse-to-fine masking: per example a level $l \sim \mathcal{U}\{1..8\}$ and a ratio $t \sim \mathcal{U}(0, 1]$; levels below l stay visible, level l is masked i.i.d. with probability t , levels above l are entirely MASK; the loss is cross-entropy on the masked cells of level l only. A phoneme prefix and a 3-second prompt of codec tokens are always visible; there is no speaker encoder, so

the backbone itself is the only identity pathway, and no classifier-free guidance, which would add a second inference-time axis and confound the step axis. The target length is fixed before generation from the text length and a corpus character rate (0.0602 s/char).

Frozen sampler. MaskGIT-style confidence decoding [Chang et al., 2022], level by level, with exactly T refinement steps per level, so $\text{NFE} = 8T$: a cosine schedule sets how many cells remain masked after each step, tokens are drawn categorically at temperature 1.0, and confidences are log-probabilities plus Gumbel noise annealed $1 \rightarrow 0$ within a level; committed cells stay committed. The sampler is identical for every config, seed and T — nothing is tuned per shape or per step budget. Per-item generation noise is keyed so that each item sees the same noise at the first step of every level for every T , making T the only difference between surface points of one run. Before any metric is computed we verify that T reaches the sampler: outputs at $T=1$ and $T=16$ must differ on more than 20% of generated token cells.

Design and statistics. 15 configs span three iso- N budgets $\times$ five aspect ratios, each trained at 3 seeds for 30,000 steps at effective batch 256 with μP [Yang et al., 2021] (base width 256, base LR 0.004); width transfer and depth transfer were verified by five-point LR sweeps before the grid. $T \in \{1, 2, 4, 8, 16\}$ is swept at evaluation only, giving 75 surface points per metric from 45 trained models. Metrics: WER (Whisper-large-v3 [Radford et al., 2023], greedy, the Whisper English normaliser), SIM-o (WavLM-large [Chen et al., 2022] speaker-verification embedding against the *original* prompt waveform, following Anastassiou et al. [2024]), a frozen degenerate-output rule, and UTMOS [Saeki et al., 2022] as a tertiary. Error transforms are $\text{err}_{\text{WER}} = \text{WER}$ and $\text{err}_{\text{SIM}} = 1 - \text{SIM-o}$, computed over all items including degenerates. Both forms of Eq. (1) are fitted by weighted non-linear least squares with 32 multi-starts; $\Delta\tau = \tau_{\text{WER}} - \tau_{\text{SIM}}$, κ , and $\Delta\rho = \alpha_{\text{SIM}}/\beta_{\text{SIM}} - \alpha_{\text{WER}}/\beta_{\text{WER}}$ carry percentile confidence intervals from a bootstrap that resamples at the *run* level — the five T -points of one run share a trained model, so resampling surface points would understate their correlation — with 2000 replicates recomputing every T point.

4 Results

Steps are metric-selective (H-D2, supported). Fitted on all 75 surface points, the step exponent is $\tau_{\text{WER}} = 0.836$ (CI [0.825, 0.851]) against $\tau_{\text{SIM}} = 0.726$ (CI [0.709, 0.744]), so $\Delta\tau = 0.110$ with CI [0.092, 0.131], excluding zero. The raw surface shows it without any model: over the grid, $T=1 \rightarrow T=16$ takes WER from 113.8% to 18.7% (a factor of $6.1\times$ on the error scale) while SIM-o moves only $0.209 \rightarrow 0.370$ ($1.25\times$ on $1 - \text{SIM-o}$). The power gate passed in all three budgets before these tests were run.

Depth and steps do not substitute (H-D3, not supported). The pre-registered substitution form collapses: $\Delta\text{AICc}(M_{\text{sub}} - M_{\text{sep}}) = +69.3$ for WER and $+71.3$ for SIM-o, where the hypothesis required $\leq +4$, and the fitted κ sits on its bound for both metrics, i.e. the exchange rate is not identified. We therefore quote no κ . The mechanism is visible in the fits: T enters with a large additive term that decays as $T^{-\tau}$ towards a floor, whereas depth changes the floor itself; a model that forces the two into one factor dT^κ cannot fit both the strong depth ordering at fixed T and the strong T ordering at fixed depth.

Shape matters, but the anisotropy ratio is not identified (H-D1, not supported). At $T=16$ the two-axis model beats the N -only model for both metrics ($\Delta\text{AICc} = -15.5$ for WER, -5.5 for SIM-o, against a -4 threshold), so error is not a function of parameter count alone — iso- N shape choice matters. The direction of the anisotropy, however, we cannot claim: $\Delta\rho = 0.174$ with CI [-0.711, 0.388], spanning zero, and the width amplitude A saturates the pre-registered bound $A \leq 10$ in every fit, so only $A w^{-\alpha}$ is identified and α — hence $\rho = \alpha/\beta$ — is not. The *depth* exponent, by contrast, *is* identified (B interior in both fits) and separates the capabilities sharply: $\beta_{\text{WER}} = 2.040$ against $\beta_{\text{SIM}} = 0.553$. What the data do show unambiguously is a strong *depth* ordering at fixed parameter count: moving from the shallowest shape to the best-performing depth is worth 40.2, 16.3 and 7.3 WER points in the 20, 50 and 125 M budgets, while SIM-o varies by under 0.065 across all 15 shapes. The depth optimum is *interior* only in the smallest budget ($d=18$ of 4–24); in the two larger budgets the argmin sits at the deepest shape tested ($d=30$ of 6–30 and $d=36$ of 8–36), so those are censored lower bounds and we claim no $d^*(N)$ trend.

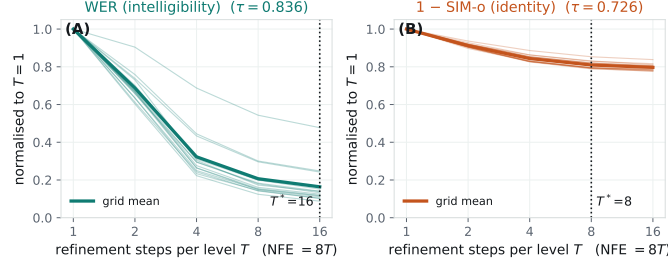


Figure 2: *Test-time scaling is metric-selective.* Error versus **refinement steps** for all 15 configs, each normalised to its own $T=1$ value; thick lines are grid means. (A) WER error falls 0.9518 in absolute terms over $T=1 \rightarrow 16$ and is still falling at $T=16$. (B) SIM-o error falls 0.1606 over the same range, $5.9\times$ less. Same models, same frozen sampler, same items — only T differs.

Table 1: *Fitted exponents* on the 75-point surface (M_{sep} for α, β, τ ; M_{sub} for κ) with 95% run-level bootstrap CIs for the step exponents. $\Delta\tau = 0.110$, CI [0.092, 0.131], excludes zero: the pre-registered primary result. κ is at its bound in both rows and $\Delta\text{AICc}_{\text{sub-sep}} \gg +4$, so no exchange rate is quotable. $\Delta\text{AICc}_{\text{full-N}} \leq -4$ means iso- N shape matters.

Metric	α (width)	β (depth)	τ (steps)	κ	$\Delta\text{AICc}_{\text{sub-sep}}$	$\Delta\text{AICc}_{\text{full-N}}$
WER	1.574	2.040	0.836 [0.825, 0.851]	3.000 [†]	+69.3	−15.5
SIM-o	0.919	0.553	0.726 [0.709, 0.744]	3.000 [†]	+71.3	−5.5

[†]at the fitted bound, i.e. not identified; $\Delta\tau = 0.110$, CI [0.092, 0.131].

Extrapolation and reliability. Fitting the two smaller budgets and predicting the largest gives MAPE 2.4% for SIM-o (versus 2.2% for the N -only model), inside the 15% the hypothesis asked for; WER does not extrapolate (20.2% versus 20.4%). Degenerate outputs — empty, truncated or looping transcripts under the frozen rule — fall monotonically with the step budget, from 40.4% at $T=1$ to 0.1% at $T=16$: most of what $T=1$ costs is not subtle quality but outright failures.

Serving corollary. Measured on the evaluation GPU at batch 1, c_{layer} is flat in width across the whole grid (0.493–0.514 ms per layer for $w \in [256, 1152]$): at these sizes the device is launch-bound, so serial latency is $8Td c_{\text{layer}}$ and width is free, so a latency budget constrains only the product Td . These data do not resolve how to split it: the pre-registered depth-first allocation rule failed (depth-first at 5.9% of 40 log-spaced budgets by the measured method and 8.8% by the fitted one, against an 80% bar), and T is censored at 16 by the tested grid, so no directional allocation rule is claimed in either direction.

5 Limitations and outlook

The ceiling is 133 M non-embedding parameters and 30,000 steps; every model is under-trained by frontier standards, and the T curves may flatten differently at scale — $T_{\text{WER}}^* = 16$ is a lower bound, since WER had not saturated at the largest budget we measured. One language, one codec, one corpus, and no classifier-free guidance — guidance is a second inference-time axis that would confound the step axis, so excluding it is deliberate, and it is the obvious follow-up. The $\Delta\tau$ interval resamples 3 runs per config, so it captures seed-to-seed variation only, not the uncertainty of the single shared harness and corpus. $\Delta\rho$ and κ are reported as unidentified rather than estimated. Cuts taken: none. This is pre-registered outcome class S1, and selectivity is the only positive claim the data license.

Reproducibility. Protocol, hypotheses, thresholds, frozen sampler and analysis plan were registered before the first run; every gate is logged with its measured values. The released artefact holds the harness, all 45 run records, the 75-point surface, and the fitting and bootstrap code. Total cost 143 GPU-hours.

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201 A The shape grid, as run

202 Every config in the active grid with its measured $T=16$ means (seeds 0, 1, 2, 400 eval_zs items each).
 203 Iso- N within a budget: the three budgets sit at 20, 50 and 125 M non-embedding parameters and the
 204 five shapes per budget span the aspect-ratio range at fixed $N = 12 d w^2$.

config	budget	w	d	heads	N (M)	val loss	WER (%)	SIM-o	Degen (%)
A1	A	640	4	10	19.7	4.638	53.9	0.327	0.67
A2	A	448	8	7	19.3	4.582	28.4	0.348	0.08
A3	A	384	12	6	21.2	4.566	15.9	0.360	0.17
A4	A	320	18	5	22.1	4.574	13.6	0.359	0.00
A5	A	256	24	4	18.9	4.611	15.3	0.351	0.17
B1	B	832	6	13	49.8	4.551	27.8	0.361	0.08
B2	B	640	10	10	49.2	4.505	16.0	0.381	0.00
B3	B	512	16	8	50.3	4.497	11.9	0.388	0.08
B4	B	448	22	7	53.0	4.511	12.0	0.382	0.08
B5	B	384	30	6	53.1	4.520	11.5	0.375	0.08
C1	C	1152	8	18	127.4	4.504	17.1	0.373	0.17
C2	C	960	12	15	132.7	4.504	18.9	0.373	0.17
C3	C	768	18	12	127.4	4.488	14.3	0.392	0.00
C4	C	640	26	10	127.8	4.489	13.4	0.390	0.08
C5	C	512	36	8	113.2	4.487	9.9	0.388	0.00

206 B Additional figures

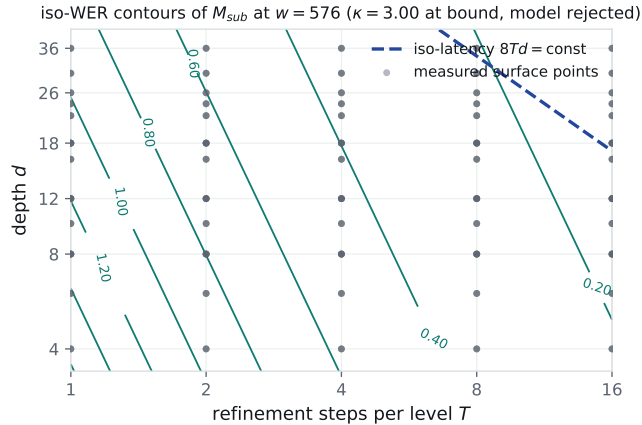


Figure 3: *Why the substitution form is rejected.* Iso-WER contours of M_{sub} in $(\log T, \log d)$ at the median width, with the iso-latency line $8Td = \text{const}$. The fitted κ sits on its bound and $\Delta \text{AICc}(M_{\text{sub}} - M_{\text{sep}}) = +69.3$, so these contours are shown as a rejected model, not as a design rule.

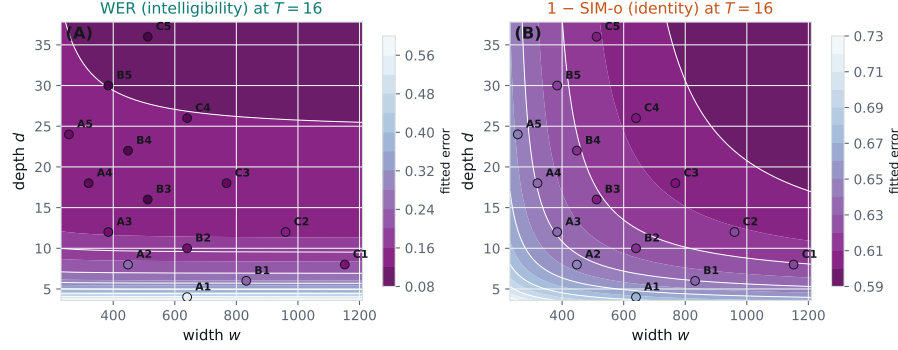


Figure 4: *Fitted anisotropy surface at $T=16$* for both metrics, with the measured config means overlaid. The width amplitude A saturates its pre-registered upper bound for both metrics, which is why $\rho = \alpha/\beta$ is reported as unidentified in §4 even though the two-axis model beats the N -only model ($\Delta\text{AICc} = -15.5$ and -5.5).

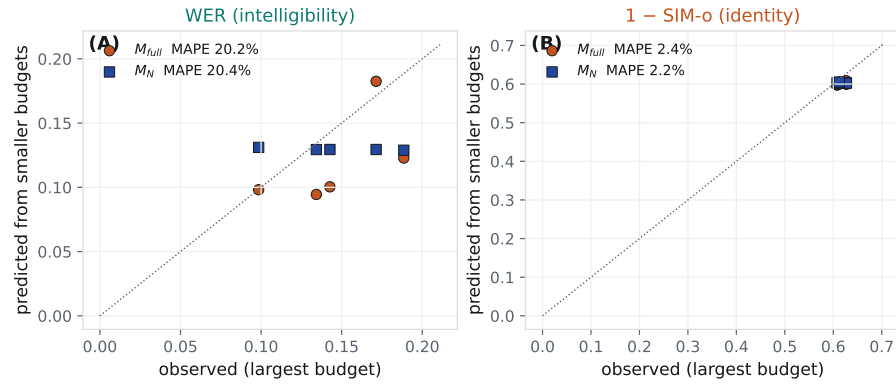


Figure 5: *H-D4 extrapolation*. The two smaller budgets are fitted and the largest budget predicted, under M_{full} and the N -only model. SIM-o extrapolates (MAPE 2.4 % vs 2.2 %); WER does not (20.2 % vs 20.4 %).

207 C Fit diagnostics and exploratory analyses

208 **Parameters at bounds.** The width amplitude A saturates the pre-registered *upper* bound $A \leq 10$ in
 209 every fit (Part A: 10.000 for both metrics; Part B: 9.999 for WER, 10.000 for SIM-o), so only the
 210 product $A w^{-\alpha}$ is identified and α is not; the bound, not the data, is binding, which is a limitation
 211 of the pre-registered parameterisation worth fixing in any follow-up. κ reaches its upper bound in
 212 every one of the 2000 bootstrap replicates of M_{sub} . The irreducible term E reaches 0 for WER under
 213 M_{sep} . The depth amplitude B is interior throughout, so β is identified.

214 **Exploratory: the depth exponent separates the capabilities.** $\beta_{\text{WER}} = 2.040$ against $\beta_{\text{SIM}} =$
 215 0.553 , a factor of 3.7 — the asymmetry H-D1 sought, in the parameter the data pin down. Part-B ρ
 216 values point the pre-registered way ($\Delta\rho = +0.890$) but inherit the A -cap problem; the pre-registered
 217 test is Part A, so this is exploratory only.

218 **Weighting.** Points are weighted $1/\text{SE}^2$ with SE taken over seeds, floored at a quarter of the pooled
 219 seed SD so that a chance agreement between two seeds cannot dominate the fit; the number of floored
 220 points is recorded in the released `fits.json`.

221 **Exploratory (not part of any declared claim).** An unweighted refit of both parts is released
 222 alongside the weighted fit, as is a fit that additionally includes the finished runs of a stretch third seed.
 223 Both reproduce the sign and significance of $\Delta\tau$.

224 **Use of Generative AI Disclosure**

225 Generative AI was used as a coding assistant for the experimental harness and for language refinement
226 of author-written text. The research design, the pre-registered hypotheses and thresholds, the analysis
227 plan, and all scientific claims are the authors' own; the authors verified the code and the numbers and
228 take full responsibility for this paper.